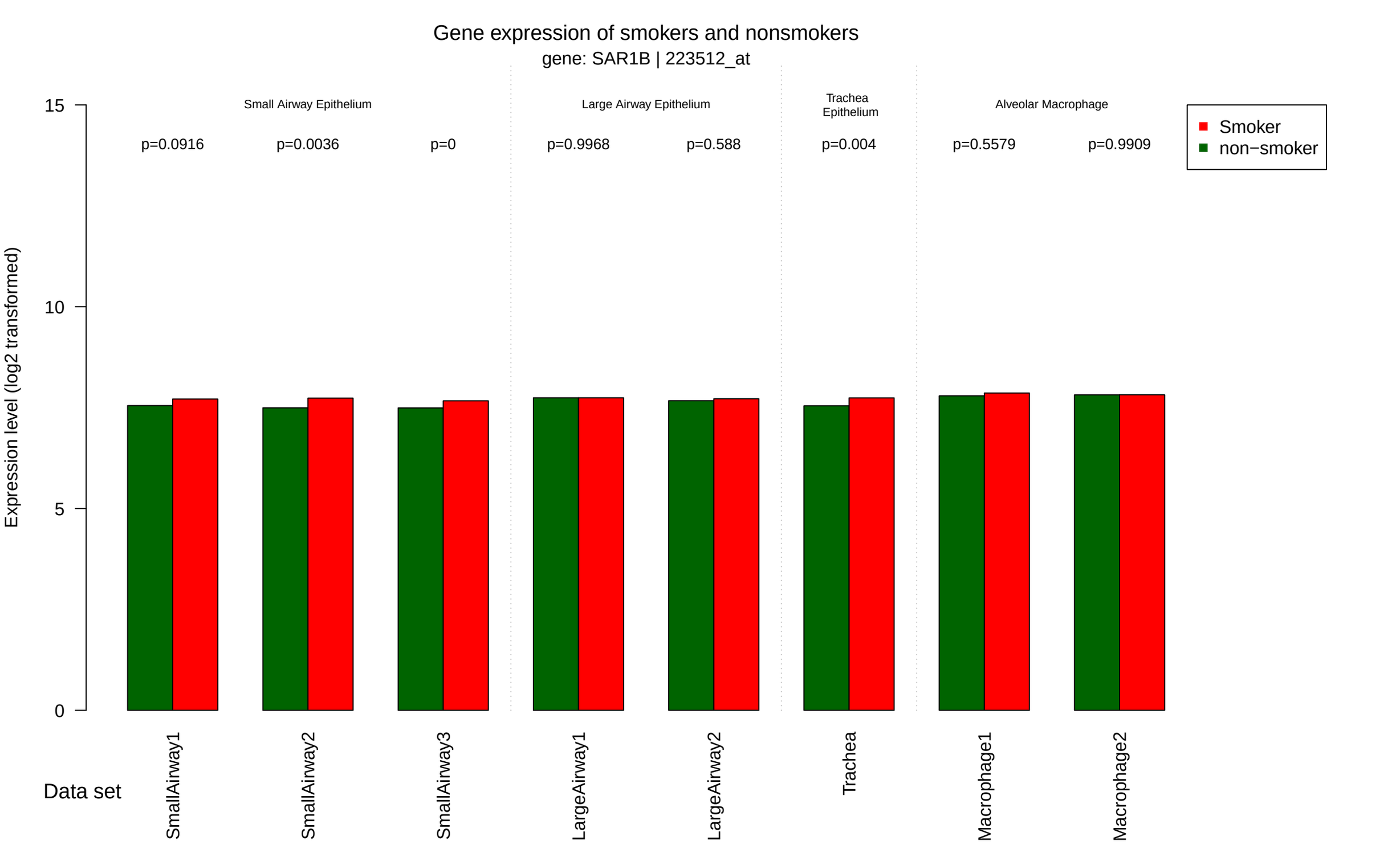
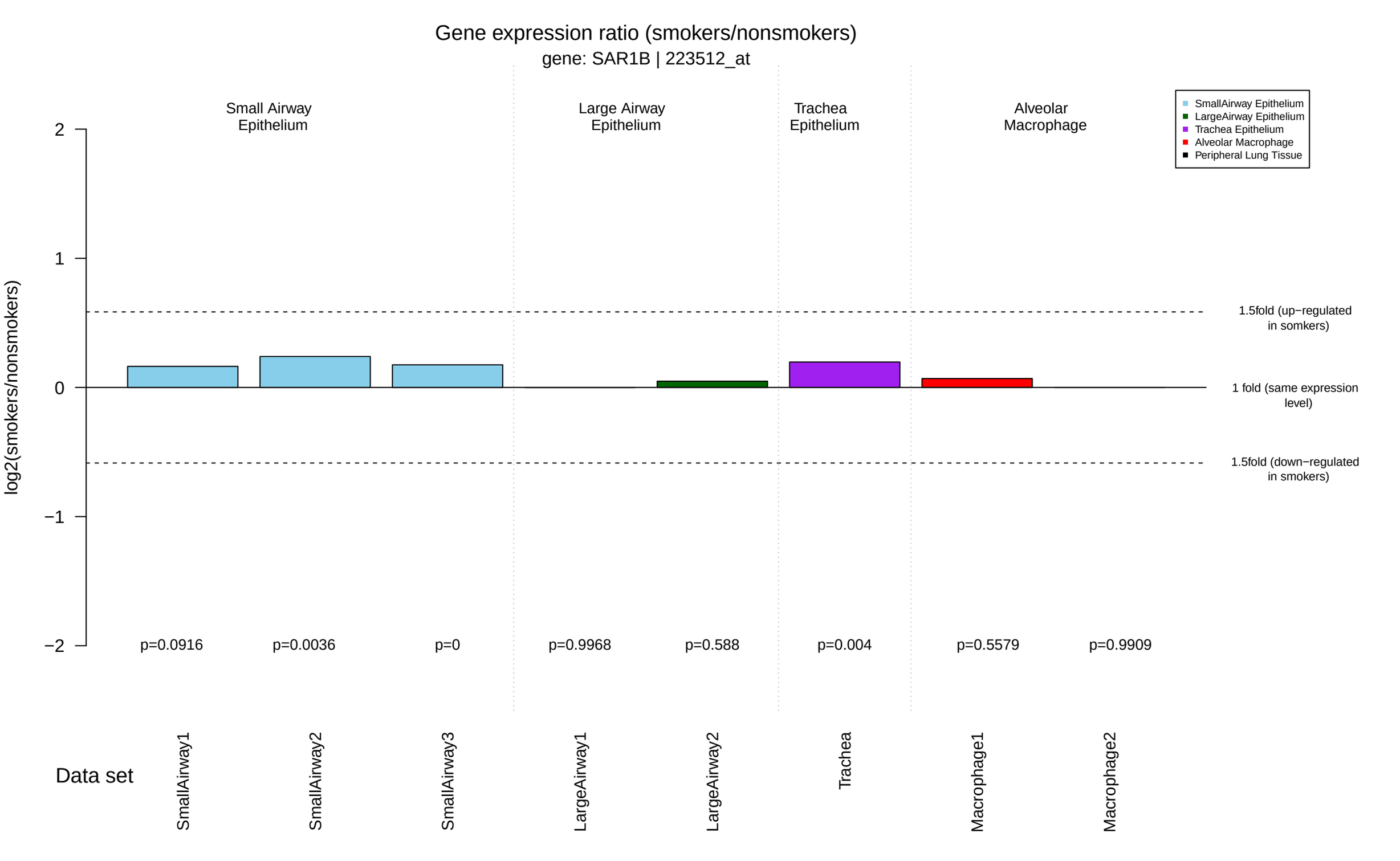
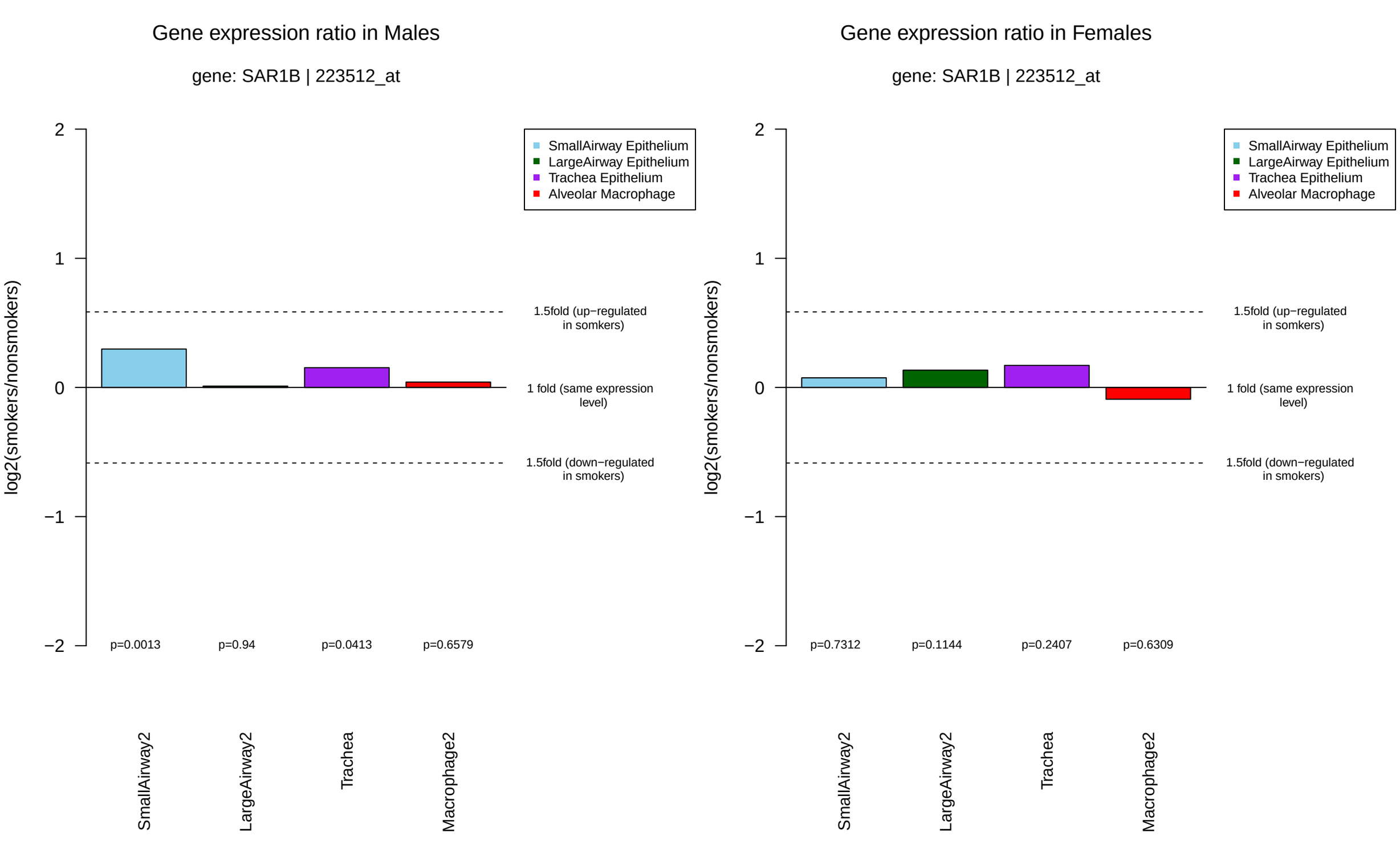
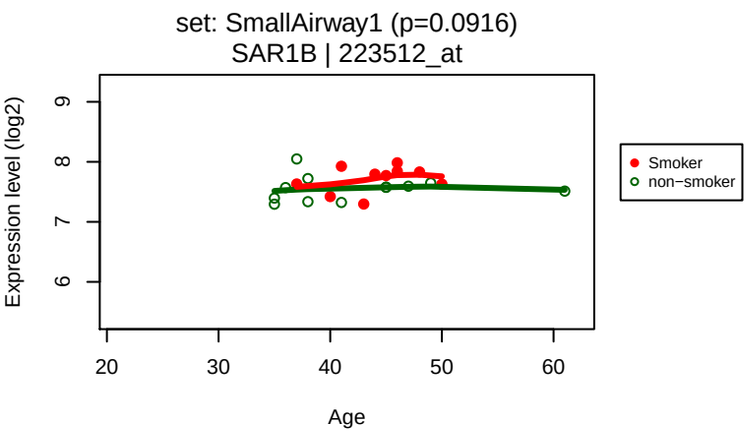
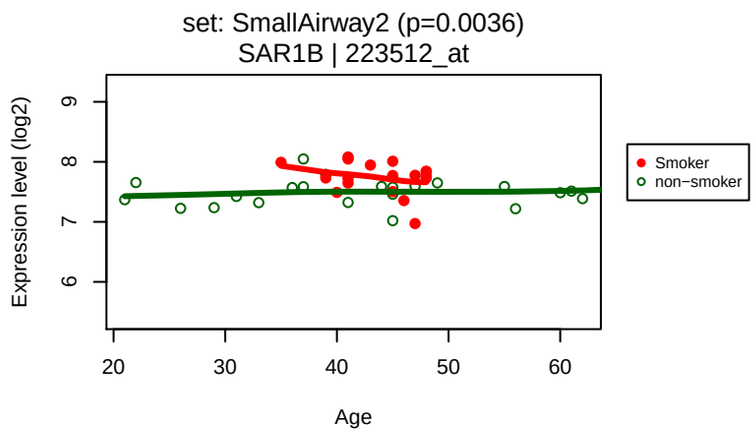
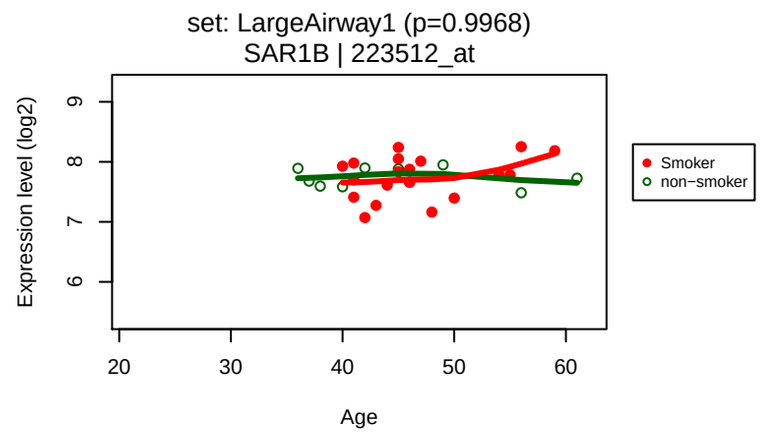

Gene expression of smokers and nonsmokers
gene: SAR1B | 223512_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0916
p=0.0036
p=0
p=0.9968
p=0.588
p=0.004
p=0.5579
p=0.9909
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

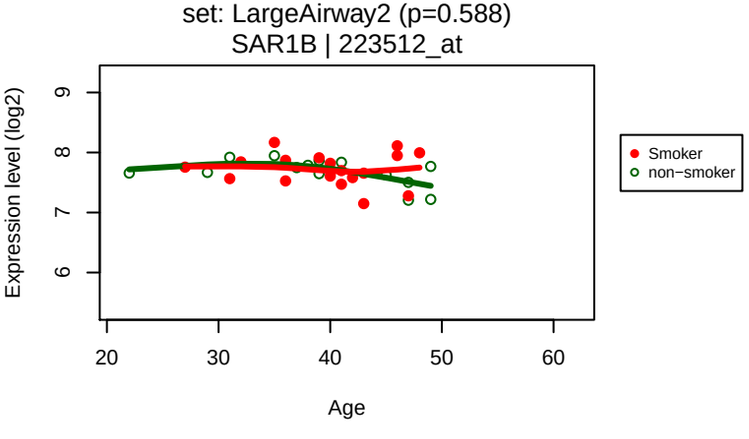

Gene expression ratio (smokers/nonsmokers)
gene: SAR1B | 223512_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0916
p=0.0036
p=0
p=0.9968
p=0.588
p=0.004
p=0.5579
p=0.9909
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

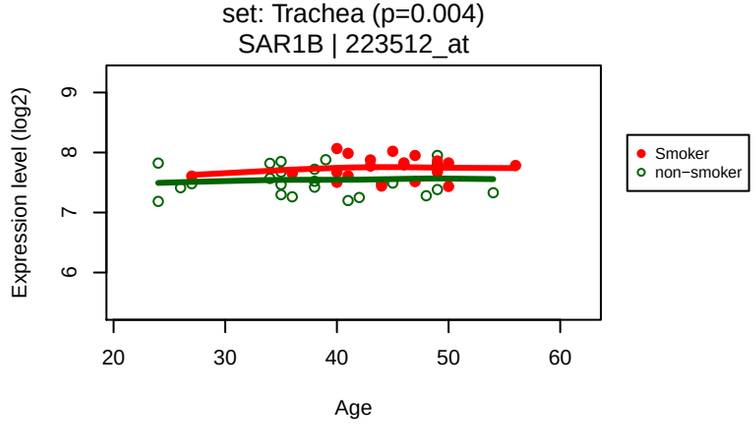

Gene expression ratio in Males
gene: SAR1B | 223512_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0013
p=0.94
p=0.0413
p=0.6579
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SAR1B | 223512_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7312
p=0.1144
p=0.2407
p=0.6309
SmallAirway2
LargeAirway2
Trachea
Macrophage2

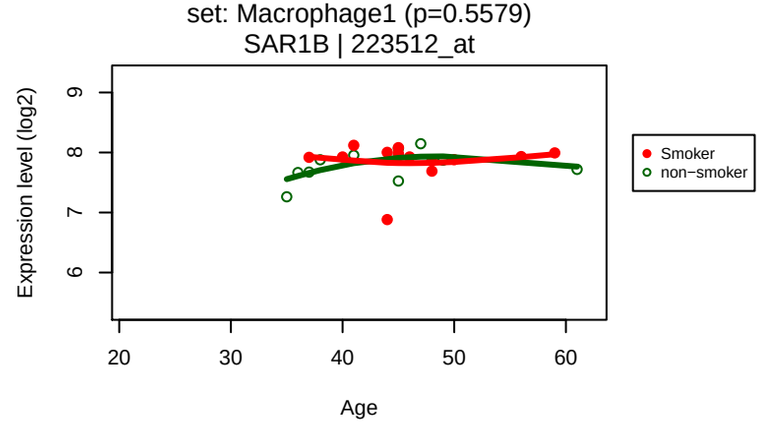

set: SmallAirway1 (p=0.0916)
SAR1B | 223512_at
Expression level (log2)
Age
Smoker
non−smoker

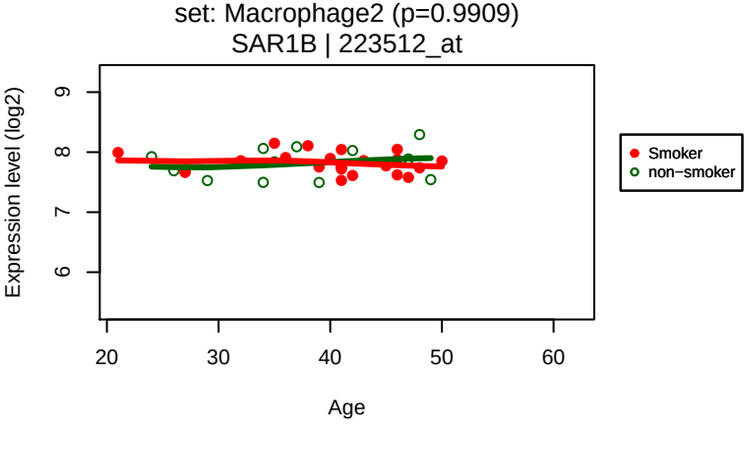

set: SmallAirway2 (p=0.0036)
SAR1B | 223512_at
Expression level (log2)
Age
Smoker
non−smoker

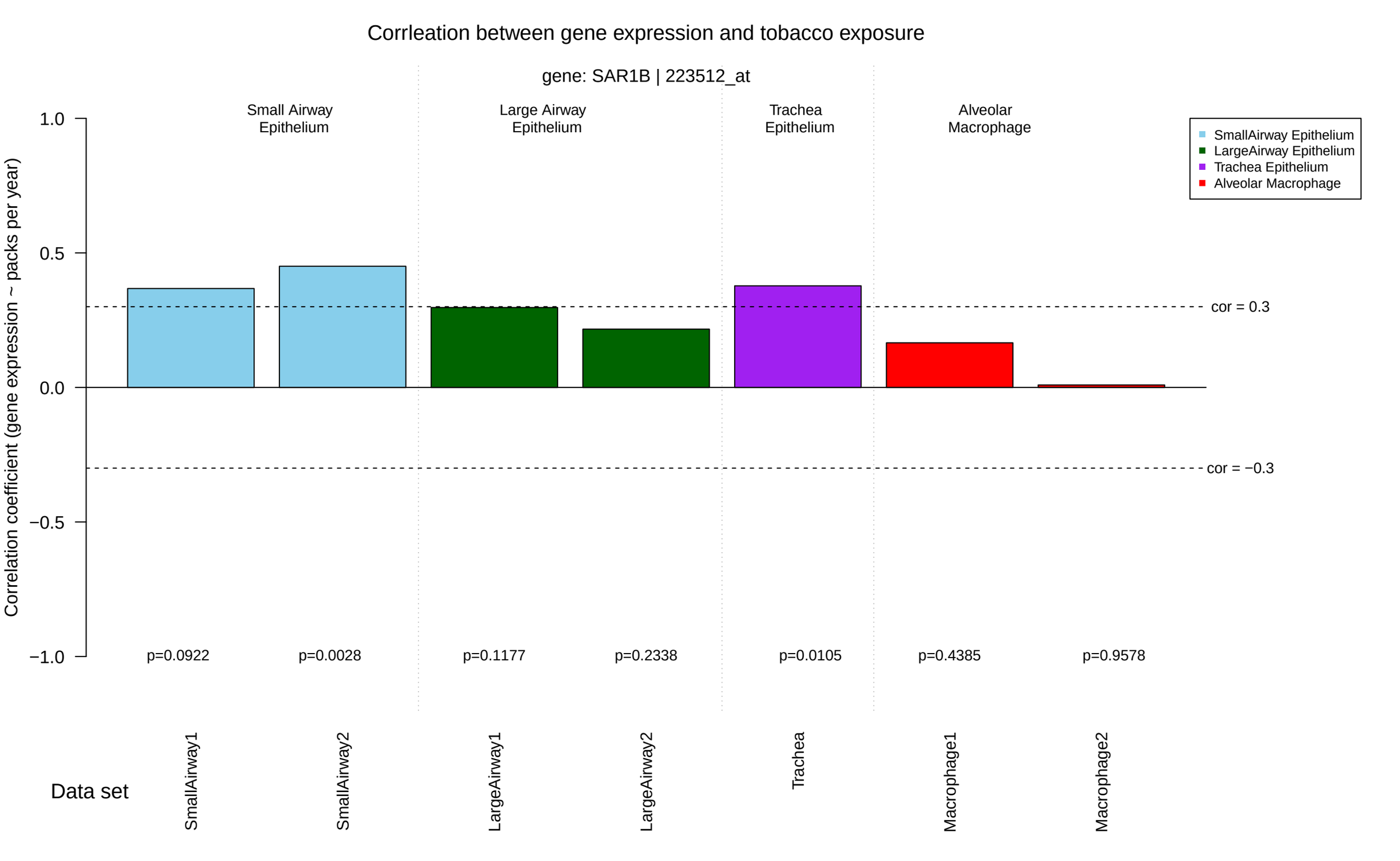

set: LargeAirway1 (p=0.9968)
SAR1B | 223512_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.588)
SAR1B | 223512_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.004)
SAR1B | 223512_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5579)
SAR1B | 223512_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9909)
SAR1B | 223512_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SAR1B | 223512_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0922
p=0.0028
p=0.1177
p=0.2338
p=0.0105
p=0.4385
p=0.9578
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2